

COMPUTER VISION USING OPEN CV – CSA 0608

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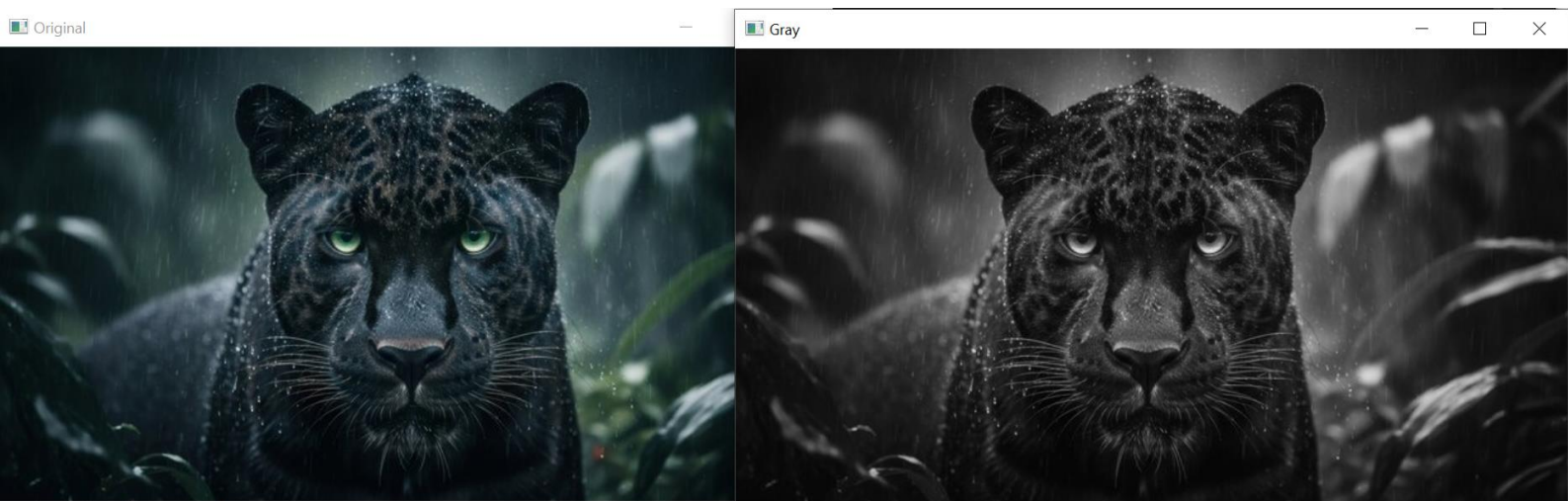
Computer Vision with Open CV LAB Experiments

1. Perform basic Image Handling and processing operations on the image.
 - a. Read an image in python and Convert the given Image into Grayscale



OUTPUT

```
main.py x
1  import cv2
2
3  img = cv2.imread("BLACK JAQUAR.jpg")
4  print(img)
5  gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
6
7  cv2.imshow( winname: "Original", img)
8  cv2.imshow( winname: "Gray", gray)
9
10 cv2.waitKey(0)
11 cv2.destroyAllWindows()
```



b. Read an image in python and Convert an Image to Blur using GaussianBlur.



OUTPUT77

```
import cv2|

img = cv2.imread("Scenary.png")

blur = cv2.GaussianBlur(img, (5, 5), 0)

cv2.imshow("Original", img)
cv2.imshow("Blurred", blur)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



c. Read an image in python and Convert the given Image to show outline using Canny function.



OUTPUT

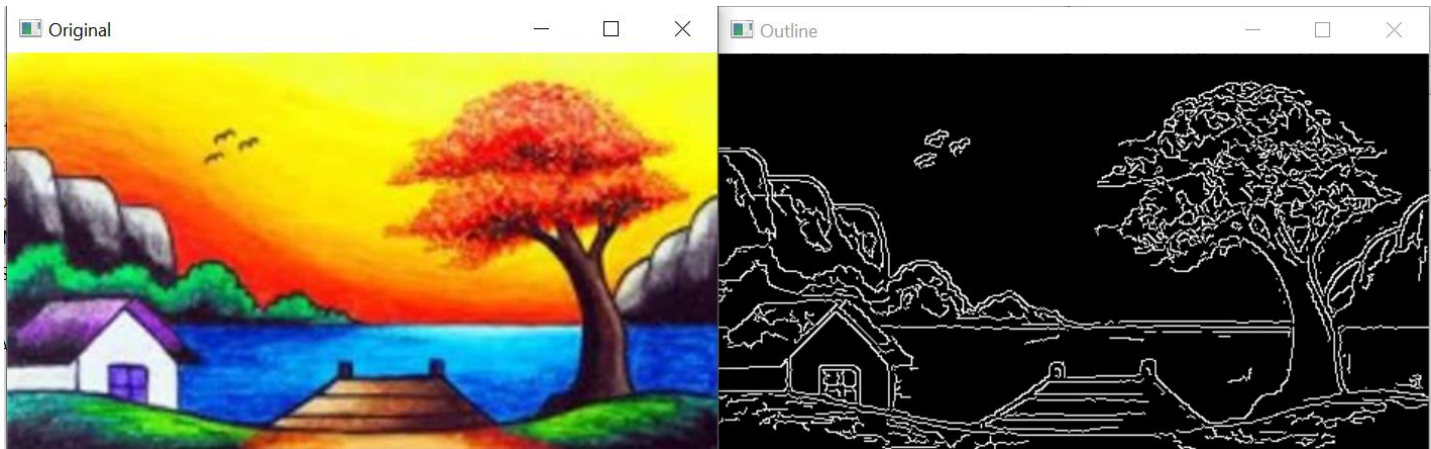
```
import cv2

img = cv2.imread("EXP3IMAGE.JPG")

edges = cv2.Canny(img, 100, 200)

cv2.imshow("Original", img)
cv2.imshow("Outline", edges)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



d. Read an image in python and Dilate an Image using Dilate function.



OUTPUT

```
import cv2
import numpy as np

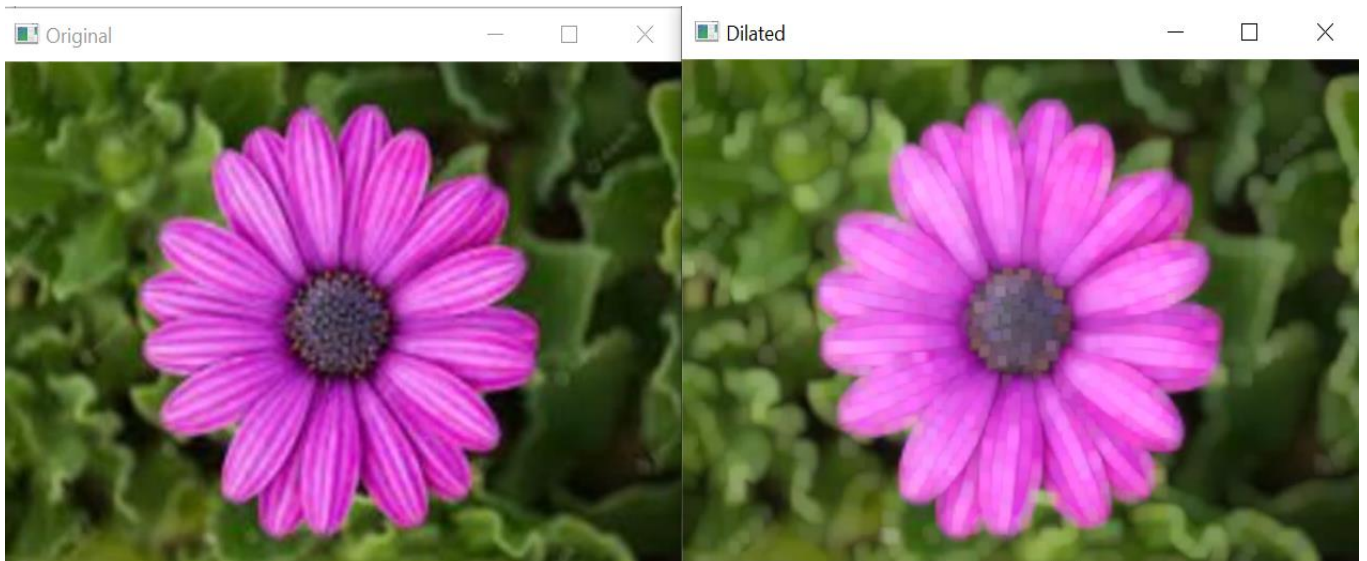
img = cv2.imread("EXP4IMAGE.JPG")

kernel = np.ones((5, 5), np.uint8)

dilate = cv2.dilate(img, kernel, iterations=1)

cv2.imshow("Original", img)
cv2.imshow("Dilated", dilate)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



e. Read an image in python and Erode an Image using erode function.



OUTPUT

```
import cv2
import numpy as np

# Read the image
img = cv2.imread("EXPSIMAGE.JPG")

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Erode the image
eroded = cv2.erode(img, kernel, iterations=1)

# Display original image
cv2.imshow("Original Image", img)

# Display eroded image
cv2.imshow("Eroded Image", eroded)

# Wait until any key is pressed
cv2.waitKey(0)

# Close all image windows
cv2.destroyAllWindows()
```

